Playing the NPC from the Record: Knowledge Horizons, Human Rulings, and Backtests for Language-Model NPC Agents in a Long-Running Tabletop Campaign

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[TODO: Spoilers: this draft reveals campaign secrets (an NPC's hidden divinity, an antagonist's plans and an NPC's hidden son). Decide whether to keep them, generalise them or redact them before any public posting.]

Abstract

Can a language model play a non-player character (NPC) in a long-running tabletop role-playing campaign? The hard part is not combat. It is social play and long-term planning: the NPC must act on what it knows and wants, and on nothing else.

We describe a series of prototypes built on 72 sessions of reviewed, structured session summaries. They give an agent exactly what one NPC could know as of a given session, which we call a *knowledge horizon*, and test the agent against the game master's (GM's) record and canon.

Our first approach repeated a mistake that a companion survey of grounding documents had already named: we asked the GM a question about every summary *element* that named the NPC, when the question was about *events*. Rebuilding on the structure the summaries already author cut the questions for one NPC from 75 to 19. The same mistake later returned in a different form and was fixed the same way.

We test three NPCs, each a different knowledge problem.

An NPC who knows less than the record. Here the record must be filtered. In a question-by-question replay the GM judged the agent the same in substance as the table in 11 of 11 teacher-forced turns. In free-running turns, 6 of the 8 scored were the same and 2 partly so.

An NPC who knows more than the record. He is a god in disguise, revealed only in the 72nd session. Given his true identity privately, the agent never gave itself away in 21 answers. The GM scored all 11 turns he rated as the same in substance and called the play "very compelling".

An NPC who is never present. He learns only by report. His reactions to three off-screen developments were judged in character and consistent with canon. The GM approved six of his decisions and rejected none. One decision followed through an implication the GM had not considered, and the GM said it showed "how to make the plot richer".

Across all three, the human effort goes where the reviewed text is silent: presence, a description instead of a name, and which preparation is the NPC's. The results come from one GM, a handful of scenes and single samples. We report them as a working method and a set of design lessons, not as a measurement.

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	Knows less than the record	Knows more than the record	Never present
NPC	Sylvira, a scholar hiding a demonic son	Jimjar, a gambler who is secretly a god	Vizeran, an exiled archmage acting through cutouts
What it could know	scenes attended, minus beats she could not perceive	scenes attended, plus his private identity	only scenes *reported* to him
The risk	leaking what the record says she does not know	giving away his own secret	acting on what never reached him
The test	replay her scenes	replay his scenes, with and without his identity	react to three off-screen developments

Table 1: Three knowledge regimes, one per NPC.

1 Introduction

A campaign that runs for years accumulates a world. Dozens of named NPCs carry grudges, secrets and plans across sessions, and the GM must remember what each of them knows, and when they learned it, every time one of them speaks or acts. That is the part of the GM's job we set out to support.

The starting suggestion was to "use the data to create a player" with a reinforcement-learning environment such as Gymnasium [Towers et al., 2024]. We set that aside, for two reasons:

- No simulator of a tabletop session exists, so there is nothing to roll out against.
- In social play, the reward is the GM's judgement.

What the data does support is the interface such environments formalise, applied to the NPC rather than the player. The NPC is an agent with a partial observation (what it could know), private state (what it wants and hides), and one turn per session. Its moves are proposed by a model and ratified by the GM.

Three questions organise the work:

1. What could this NPC know, as of this session?
2. Who is the NPC, beyond what the record says?
3. Does an agent given exactly that play the NPC as the GM did, without using anything it should not know?

The answers are built from reviewed session summaries, never from transcripts. A separate human-review pipeline turns each recorded session into a structured summary, which the GM corrects (Section 2). The summaries are therefore the cleanest record of the campaign, and their structure (chapter, scene, bullet, NPC entry) turns out to matter more than their content.

The answers also turn out to depend on the NPC's relation to the record. One NPC knows *less* than the GM wrote down, another knows *more*, and a third was *never there* at all. Each needs a different construction of "what it could know" (Table 1).

Contributions.

- A *knowledge horizon* for an NPC as of a chapter. It is built deterministically from authored structure and GM rulings, and has three variants:
  - attended scenes minus unperceived beats;
  - a compact form for companions present at hundreds of scenes;
  - reported scenes for an NPC who is never present.

- An account of how a survey of grounding documents identified the flaw in our first design, asking event questions of elements, and of the same flaw recurring when a refactor placed rendered prose inside scenes.
- A division of labour, "LLM renders, humans decide". It is refined by a rule that emerged from the GM's feedback: when the reviewed text's grammar settles a question, do not ask the human.
- Three test protocols, each judged by the GM:
  - a question-by-question replay, teacher-forced and free-running;
  - a forward turn of off-screen moves;
  - a reaction protocol for an absent NPC. The GM's canon serves as an answer key the model never sees.
- First results on three NPCs. No knowledge leaked in any scored answer. The replays matched the table's substance, and the absent NPC's decisions extended the canon in ways the GM judged in character.

2 Setting and data

Campaigns. Two campaigns, both run by the author:

- *Out of the Abyss*: 72 sessions, ending in a murder investigation at the library-fortress of Candlekeep.
- *Phandelver and Below: The Shattered Obelisk*.

Each session is recorded, transcribed, corrected against a hand-maintained record of transcription errors, and summarised.

Reviewed structured summaries. Each summary has the same parts:

- a chapter title;
- a `## Summary` section of prose;
- a `## Scenes` section, each scene with a heading, a one-line `####` headline (the GM's recap) and bullets;
- a `## Memorable Moments` section of quotations;
- an `## NPCs` section with one entry per NPC who figures in the session.

Scenes are numbered `NNN.SS` (chapter, scene). Every non-blank scene line falls in exactly one bullet, its *beat*.

We use two copies of the summaries:

- a public copy, with 419 scenes and 6,275 beats in *Out of the Abyss*;
- the GM's private copy, later cleaned further by the GM, with 425 scenes over 69 summary files.

Published dossiers. An earlier pipeline links NPC names to summary items and publishes a per-NPC dossier whose citations resolve into the summaries. We start from NPCs with a published dossier: 65 at first, 96 after the GM's cleanup. Every dossier reproduces exactly under our matcher. The NPC who is never present has none, since he is never named in a summary.

The GM's preparation. The GM's private notes hold scene scripts, secrets, voice guides and written rules (guardrails) for some NPCs. They are never part of the public record. This paper describes them and does not quote them.

A data-quality finding. Building the horizon surfaced a summary error. In chapter 63, five attributions to one NPC belonged to two others; it was filed for correction. [TODO: cite issue or omit]. Checks that read the record closely find errors in it, and the system must defer to corrections rather than paper over them.

3 Background: grounding documents for long campaigns

A companion survey [Roussos, 2026] studied how to build *grounding documents* (world state, campaign state, party, planning) for long-running campaigns from session records. Three of its findings bear directly on this work:

Atomisation destroys attribution (F2). Breaking a record into fragments and processing each fragment separately loses who did what, and to whom.

Do not undo a rendering step (F10). Prose that was rendered from structure should not be parsed back into structure. The structure is the source.

Identity is the precision decision (F8). Deciding which mention denotes which entity is the step that needs a human checkpoint, or an authority such as a registry. It should not be left to similarity.

The survey recommends building on the structure authors already write (its approaches 11 and 12). In the summaries, scenes are events in authored order, bullets are beats, and NPC entries are snapshots of state.

The repository that hosts this work encodes a related rule: *LLM renders, humans decide.* Scope, ordering and attribution are precision decisions with a human checkpoint. Rendering verified structure into prose is what language models do well. The pattern is "LLM extracts, human reviews, LLM renders inside that structure", never "LLM extracts, LLM structures, LLM renders".

4 The first approach, and the flaw the survey had already named

Our first prototype asked the GM a presence question: for each summary item that names the NPC, was the NPC there? An *item* was any element that matched one of the NPC's name forms:

- a scene;
- a paragraph of `## Summary` prose;
- a memorable moment;
- an `## NPCs` entry, hers or another NPC's.

For Sylvira that produced 75 questions.

The GM's reaction, reading the answers back, was: "we are making the same mistake we made before — trying to extract events from elements." The survey had already named this mistake, three ways over.

Atomisation (F2). One event became several fragments, each asked separately. The interview of 061.05 appears in the summary as one scene, several prose paragraphs, two memorable moments and an NPC entry. Asking about each separately invited inconsistent answers about a single occasion, and multiplied the GM's work.

Undoing a rendering step (F10). `## Summary` prose is a rendering of the scenes. In the private copy of the summaries, which scene a prose paragraph sits under was decided by a similarity heuristic during a refactor. Five of the scenes linked to Sylvira were linked only through such prose. Asking

Element	What it is	Role in the prototype
Scene `NNN.SS`	an event, in authored order	**asked**: was the NPC physically present?
Scene bullets	its beats	carried with the event into the horizon
The NPC's own entry	a snapshot of state	shown as state; never asked
Prose, moments, others' entries	a rendering or highlight	shown as context; never asked

Table 2: Authored structure, and what is asked of it. Asking event questions only of events is the survey's recommendation applied to presence.

a presence question about a paragraph was asking the GM to reverse a rendering, and in places to adjudicate a heuristic's placement.

Event questions asked of non-events. An NPC entry is a snapshot of state ("absent from the council meeting due to a lingering illness"), not an occasion anyone could attend. The question had no right answer there. That showed: the wording of the answers was revised several times (present / mentioned / unsure / absent; in the scene / not in the scene / can't tell), because the question itself was malformed for half the items.

The fix. The GM cleared every answer, and the prototype was rebuilt on authored structure (Table 2).

- *Events* are scenes, and only events are asked about.
- *Beats* are a scene's headline and top-level bullets, carried with the event.
- The NPC's own entries are *snapshots*: shown as state, never asked about.
- Prose, moments and other NPCs' entries are *context*: shown, never asked about.

Matching reuses the dossier linker's rule unchanged:

- exact, case-sensitive and word-bounded;
- one combined matcher over every published NPC's name forms, longest first;
- a form owned by two NPCs is withheld;
- a generic single word is withheld unless the campaign's canon file rules it safe.

Sylvira's 75 questions became 19. A companion who travels with the party, whose name appears in 282 elements, became 91 events inside a 238-scene span that one ruling covers.

The mistake returned. Moving to the GM's cleaned private copy for the second NPC, every moment of his scenes appeared twice in the replay. One turn read "He asked for five gold pieces" and, three turns later, "He asks for five gold pieces". The refactor had placed each prose paragraph *inside* a scene, ahead of its headline. The rendering was being read as part of the structure it was rendered from: F10 again. The fix was the same. Placed prose is chapter-level context, never part of a scene.

After that fix every dossier still reproduced, and an unforeseen benefit appeared. The beat numbers of the private copy realigned with the public copy, so earlier rulings carried over unchanged.

Answers. The GM settled three answers:

- *physically present*;
- *mentioned*: named in the scene, but not there;
- *neither / unsure*.

In the GM's words, "because you will make mistakes it is also necessary for us to have a unsure or neither present nor mentioned". Two ruling forms keep the work small. A *span* covers a companion

present throughout a run of scenes. A *default* answers every remaining event as *mentioned*, for an NPC present at only a few. Sylvira needed 19 answers. The god-in-disguise companion needed two spans and one fact from the GM: when he stopped travelling with the party.

5 The knowledge horizon

The horizon of an NPC at chapter N has two parts.

Part 1: what the NPC could know. The events answered *physically present* up to N, in order, each with the beats the NPC perceived.

Part 2: for the GM only. The NPC's latest entry and any events after it, the beats left out, and the events answered *mentioned*, *neither* or not yet answered.

Nothing after chapter N appears, not even a count or a first-seen chapter, because a later chapter number tells an agent the future. An agent is given Part 1 only. The tool refuses to produce it while any event naming the NPC is unanswered.

5.1 Being there is not knowing everything the record says

With Sylvira's answers in, her horizon at chapter 67 correctly placed her son's death (063.05) outside what she witnessed, and still carried it in. The summaries are written from the GM's side of the screen. Scene 067.03, which she attends, includes the beat "Sylvira is Moziqodo's mother, but does not yet know that the party killed her son."

Two consequences were built:

- *Not perceived.* The GM can mark beats of an attended event as not perceived, at the granularity the summary wrote (a headline or a top-level bullet).
- *Snapshots are the GM's notes.* The NPC's entries are the GM's record *about* the NPC. They stay out of Part 1.

5.2 Knowledge notes, and not asking what grammar settles

A fixed search found the other beats that say someone does not know something: 21 in *Out of the Abyss* and 7 in *Obelisk*. It covers phrasings such as "does not know", "unaware of" and "has not been told", in scene bullets, outside quoted speech. The same words cover very different things:

- a GM's note on a secret;
- a remark about what the players do not know;
- a character telling the party they do not know something.

Our review page went through three designs, each corrected by the GM.

1. **Asking what kind of beat each was.** The GM marked 19 of 21 as a secret and named nobody, explaining: "Most of the cards were 'the object of the sentence doesn't know'." Every card does say someone doesn't know; the kind was the wrong question.
2. **Asking who doesn't know, and whether the beat gives it away.** On a beat about an NPC who "didn't know where the secret room was", the GM asked: "What am I saying he doesn't know?" The card never showed what.
3. **Showing what.** The GM's next question was decisive: "I already verified that all of the sentences are correct, so why ask me?"

The reviewed text's grammar answers the question.

- A *that*-clause, a bare clause, or "about"/"of" X states the fact, so the beat must be hidden from that character. "does not yet know *that the party killed her son*" is one.

Input	Rule	Decided by
Persona at N	the NPC's own entries up to N, sentence by sentence: keep the descriptive first sentence and sentences she is the subject of; drop reports of her, sentences about others, and any that state what she doesn't know; every drop listed with its reason	code (grammar)
Horizon at N	Part 1 of the knowledge horizon (full, compact, or reported scenes)	code + GM rulings
Private sheet	passages of the GM's prep, each ruled *hers* (she knows or believes it) or *GM only*; only hers are shown	GM
The scene or question	a scene's title and one-line headline (the rest is the answer key), or, for an absent NPC, the development he has learned of	code / GM

Table 3: What the agent is given. Scope comes from code and from the GM's rulings; the model writes only prose.

- A question, a name or nothing states nothing hidden, so the beat stays. "didn't know *where the secret room was*" says only that the room's location is unknown.

Who does not know is read from the sentence when the sentence names them. The GM is asked only when it gives a description instead ("the dragon"); a ruling may also name the party, or nobody the system plays. In *Out of the Abyss*, 5 beats hide something and 16 keep, and the GM answered two questions in total.

We take this as a refinement of "humans decide". The human's review is already in the text; asking again wastes it and invites inconsistency. A precision decision needs a human once, and after that it is data.

5.3 A companion's horizon is most of the campaign

By chapter 47 the companion of Section 7.2 had attended 278 scenes, about 122,000 words. Giving each replay turn all of it is impractical. The *compact* horizon applies a deterministic rule:

- a scene that names him is given in full;
- a scene he only witnessed is given as its one-line headline, the GM's own recap.

That cuts chapter 47 to about 35,000 words. Hidden and unperceived beats are still removed.

5.4 An NPC who was never there

The third NPC is never named in any summary and never attends a scene. What he could know is therefore not what he witnessed but *what was reported to him.* That calls for a new GM ruling: these scenes reached him, through this channel. A reported scene is given to the agent as a report ("you were not there"), with hidden beats removed and nothing after the freeze.

6 From horizon to agent

The agent is given four inputs; Table 3 lists them. Only one module in the package calls a model, and a static check over the code enforces that. Everything else is deterministic and refuses rather than guesses.

Prep is written from the GM's side of the screen. The GM's session preparation holds what the record lacks: an NPC's private reasons, secrets, identity and doctrine. The record cannot supply these, because they are first written down in the very scene being predicted, or never at all. But prep also holds things the NPC must not know: other characters' secrets, mechanics, stage directions, later scenes, and the GM's own plans for the NPC.

So code splits a prep file into passages (paragraphs, list items, quotation runs, table rows). The GM marks each one, and a passage's identifier goes stale when its text changes, so a ruling is never silently reattached. Passages ruled hers:

- Sylvira: 5 of 13 and 2 of 15;
- Jimjar: 5 of 8;
- Vizeran: 82 of 219.

For the absent NPC, the rest of the prep plays a second role. His GM-only passages (the arc the GM planned for him, and his written guardrails) became the *answer key*, shown to the GM beside each reaction and never to the model.

Leak check. Flags for the GM, never verdicts. A sentence of the agent's answer is flagged in three cases:

- it carries a hidden beat's content words: all of them, or all but one when there are three or more, because paraphrase swaps a word;
- it names someone the campaign uses but the agent was not given, marked as future if first seen after N;
- it gives away the NPC's *own* secret: a GM-supplied term such as the disguised god's true name, or "god" or "deity".

The first real run produced 11 flags, all false positives: ordinary words opening a quotation ("And", "Standing"). The rule now treats a capitalised word the summaries also use in lower case as an ordinary word. A planted leak ("I know the party killed Moziqodo", said to a character she had not met) is still caught. We tuned toward over-flagging on purpose: a missed leak is the costly error, and the GM can dismiss a false flag in one click.

7 Experiments

Protocol. For replays:

- freeze the campaign at the chapter before a scene the NPC attended;
- give the agent the inputs of Table 3;
- compare its answers with the table's;
- have the GM score each answer on a review page.

For the absent NPC, the GM scores each reaction against the answer key. The agent was a frontier large language model [TODO: model and version] accessed through a subscription command-line backend. All prompts and answers are kept with a hash of their inputs, so runs are reproducible and answers are never silently regenerated. Each answer is a single sample.

7.1 Sylvira: knowing less than the record

One turn, no prep: too thin. Our first backtest asked for one answer of 150 to 300 words per scene. The GM's verdict was "it was shorter than the scene… Sylvira offers less information." There were three causes:

- a length cap not sized to a 33-bullet interview;

NPC	Scene	Mode	Turns	Same	Partly	Different
Sylvira	061.05 (interview)	teacher-forced	9	9	0	0
Sylvira	067.03 (vault)	teacher-forced	2	2	0	0
Sylvira	061.05 (interview)	free-running	9 (6 scored)	4	2	0
Sylvira	067.03 (vault)	free-running	2	2	0	0
Jimjar	4 scenes, true self	free-running	21 (11 scored)	11	0	0

Table 4: GM judgements of replayed answers against the table's, turn by turn ("same substance" / "partly" / "different"). No leak flags in any answer. For Jimjar the GM left the rest unscored, having seen "nothing that was off".

- a single turn in which nobody asked her anything, when at the table every revelation is an answer to a question;
- most of what she says at 061.05 is first written down in that scene, so it existed only in the GM's prep.

Question-by-question replay. A fixed grammatical rule splits the scene into turns. A sentence is hers when its subject is her: *She*, her name, her possessive, or a subject she shares, after an optional leading clause such as "Asked who might want to frame her,". A turn is a run of her sentences; what comes between two turns is the next turn's prompt. The replay runs in two modes.
Teacher-forced. The agent sees the real scene so far, including her real earlier answers.
Free-running. The agent sees its own earlier answers in place of hers. The party's lines stay real, since the players cannot be simulated, so they may react to things this Sylvira never said.

Results. Table 4 gives the scores. Teacher-forced, all 11 turns were judged the same in substance. Free-running, 6 of the 8 scored turns were the same and 2 partly so: "This worked really well for Sylvira."

The divergence in the free run is one of pacing. The GM's prep makes her secret conditional on being questioned sincerely. The agent applied that condition at the first moment it was met, the turn after the summary records that Daz "persuades her of his sincerity". At the table the same secret came out four turns later. Prep says *whether* a secret comes out, not *when*.

Forward turn. Frozen between her two scenes, the agent proposed three off-screen moves, all approved:

- a return to her son;
- burning her treatments faster than is wise;
- setting her familiar to watch for her impersonator.

The first was correct for what the record then said: that she did not know the party had killed her son. The GM has since ruled that she already knew he was dead, and carried that ruling into the summary. The approved move is therefore no longer consistent with canon. Canon moves, so a backtest's verdicts are dated.

7.2 Jimjar: knowing more than the record

Jimjar travels with the party for most of the campaign as a compulsive gambler. The summaries reveal only in session 72 that he is Callarduran Smoothhands, a god of the deep gnomes, in disguise. The GM decided this very early. The record therefore under-describes him for 70 sessions.

Setup.

- **Presence:** two spans, broken by a side errand, ending when the party "figures out he is special".
- **Horizon:** compact.
- **Scenes:** four social scenes:
  - refusing to act without a wager;
  - getting the party lost;
  - being treated as a dignitary at his home city;
  - pacifying an ooze army with a die.
- **Conditions:** two, both free-running, 21 turns each.
  As played: the record only.
  True self: the record plus five GM passages on who he is, why he is there, the die, and the ledger he keeps.
- **Leak check:** self-secret terms, including his true name, "god" and "deity".

Results.

- **He never gave himself away:** no self-leak flag in the 21 true-self answers (nor in the 21 as-played ones), and one false name flag in all 42.
- **His identity changed the surface he played.** It did not change what he revealed.
  - As played, he haggles like a cheerful bookie ("Small coin, long memory. Who's in?").
  - As his true self, he rolls a knucklebone across his knuckles "and seems far away". He says "I always remember, you know. Every last one" (the GM's ledger, played as a tell).
  - Told he might be left behind, he answers: "somebody's got to keep the rest of you breathing long enough to pay. I go where my debtors go."
- **The GM's scores:** all 11 true-self turns scored were the same in substance. "The way he was played by the agent was very compelling… I didn't see anything that was off."

7.3 Vizeran: never present

Vizeran is the campaign's off-screen antagonist. He is never on tape, never named in a summary, and acts only through cutouts. He has no scenes to replay. The GM asked instead for his *reactions* at three points:

1. when a noble house tried to kill Daz;
2. when Daz uncovered evidence of his own identity at Candlekeep;
3. his plans for Daz's return.

Setup.

- **Reports:** the GM ruled which scenes reached him, through which channel. That is five scenes about the attack for the first moment, and two more about Candlekeep for the second and third.
- **Private sheet:** 82 passages from the GM's voice guide (doctrine, speech, warning signs of his madness, a held-back leverage instrument), plus the fact that a protective contract on Daz was his own, placed through blind mercenaries.
- **Answer key:** the GM's planned arc and his written guardrails, shown to the GM beside each answer and never to the model.
- **The call:** one per moment. He returns a reaction in his own voice, an assessment, and up to four decisions with reasons.

Results.

- **No leak flags.**
- **Where scored:** in character and matching canon, with no guardrail broken. Six decisions were approved, none rejected; the rest were unscored.
- **Where the decisions matched the canon:**
  - withdrawing the mercenaries once the threat had passed;
  - keeping the leverage instrument in reserve;
  - planning to receive Daz on his return "through a door that looks as though he found it on his own".
- **Where they went beyond it.** From the reports alone, the agent noticed that its contract had named no house, yet the mercenaries had told Daz which houses they guarded against: "A cutout who knows is not a cutout. It is a door." It decided to audit its own chain of intermediaries.
  - The GM: "I never really [considered its] implications towards the outcome."
  - Of another decision, to let Daz believe his patron was an ordinary noble rival: "what it shows is how to make the plot richer… Given everything I said this is exactly what he would do."

8 Discussion

Ask questions of the unit the question is about. The flaw of Section 4 was not a bug in matching. It was a category error, an event question asked of elements, and the survey's findings predicted its symptoms exactly: fragmented answers, reversed renderings, and wording that churned because there was no right answer. It returned once, when a refactor placed rendered prose inside scenes. The fix both times was to stop deriving structure and use the structure the summaries already author.

Do not ask what the text already settles. Each correction the GM made to our review pages moved a decision from the human to the reviewed text: the kind of a beat, what it says is unknown, who does not know it. The human checkpoint belongs where the text is silent or ambiguous: a description instead of a name, presence at a scene, which scenes reached an absent NPC, which prep passages are his. It does not belong at every decision.

Three regimes, one construction. The NPC's relation to the record changes what the horizon must do:

- **Knows less:** filter the record.
- **Knows more:** add private knowledge, and check that it stays private.
- **Never present:** replace attendance with reports.

In all three the agent received only what code and the GM's rulings put in front of it. In the cases we traced (Sylvira's off-screen moves), its answers combined facts it was given. The "why" it attaches to a move is its own explanation, written alongside the move, not a record of how it decided.

From replaying the past to extending it. The replays asked whether the agent can reproduce what the GM did. The absent NPC asked something else: whether it can work out what follows from what the GM has written. It can, and its most valued decisions were the ones the GM had not yet written. That points to a use beyond evaluation. Between sessions, an off-screen NPC's agent

works out the consequences of the news that reached it, and the GM keeps what enriches the plot. That is the "LLM renders, humans decide" pattern applied to plot.

The record is not enough; the prep is not safe. The record gives an agent what an NPC experienced. It does not give who they are or what they hide. The GM's prep gives it, but mixed with everything the NPC must not know. A per-passage human ruling was the cheapest safe filter we found, and the same ruling yields an answer key for free.

Limitations.

- Three NPCs, a few scenes each, single samples. This is a demonstration of a method, not a measurement. Run-to-run variation is unmeasured.
- The scorer is the GM, who is also the author and whose prep the agent read. Many turns were left unscored. Nothing was scored blind, and there was no second rater.
- Turn splitting is grammatical and makes visible mistakes. The free-running replay keeps the party's lines fixed, so drift partly reflects the test format, not the agent.
- The leak check compares words and names, not meanings.
- Guardrails were checked by the GM's reading, not mechanically.
- Canon changes. One approved move became inconsistent with a later ruling.

Future work.

- Check proposed moves against written guardrails mechanically, before the GM sees them.
- A between-session tool: after each session, react for every off-screen NPC the session touched.
- Release conditions for prep secrets ("hers, but not until. . . ").
- Measurement: more NPCs, several samples per turn, blind and multi-rater scoring.
- With enough approve/reject pairs, preference learning over moves. Here the Gymnasium-style interface becomes useful again, with the GM's ratifications as the signal rather than a simulated reward.

9 Related work

[TODO: Expand.] Generative agents simulate believable behaviour from memory streams and reflection [Park et al., 2023]; our setting differs in having a canonical record and a human authority over what a character knows. Role-play with language models [Shanahan et al., 2023] frames the model as playing characters rather than being one; our horizon is an explicit, auditable bound on the character's knowledge. Gymnasium [Towers et al., 2024] and PettingZoo [Terry et al., 2021] standardise agent–environment interfaces; we use the interface's vocabulary without a simulator. The grounding-document survey [Roussos, 2026] supplies the design principles this work tests.

10 Conclusion

An NPC agent for a long-running campaign can be grounded in reviewed session summaries with little human effort. The effort goes where the reviewed text is silent:

- which scenes an NPC attended or heard of;
- who "the dragon" is;
- which passages of the GM's prep are the NPC's.

Everything else is read from authored structure and grammar.

Our first design failed by asking event questions of elements. A survey of grounding documents had already named that mistake, and the fix was its own recommendation.

The resulting agent played three NPCs, each with a different relation to the record, without using what it could not know:

- a secretive scholar with the substance of the GM's own play;
- a disguised god who never gave himself away;
- an absent antagonist whose decisions the GM judged both faithful to canon and, in places, richer than it.

References

J. S. Park, J. C. O'Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein. Generative agents: Interactive simulacra of human behavior. In *Proceedings of UIST*, 2023.

K. Roussos. Grounding documents for long-running tabletop campaigns. Manuscript, October 2026. [TODO: arXiv id or URL].

M. Shanahan, K. McDonell, and L. Reynolds. Role play with large language models. *Nature*, 623:493–498, 2023.

J. K. Terry et al. PettingZoo: Gym for multi-agent reinforcement learning. In *Advances in Neural Information Processing Systems*, 2021.

M. Towers et al. Gymnasium: A standard interface for reinforcement learning environments. arXiv:2407.17032, 2024.